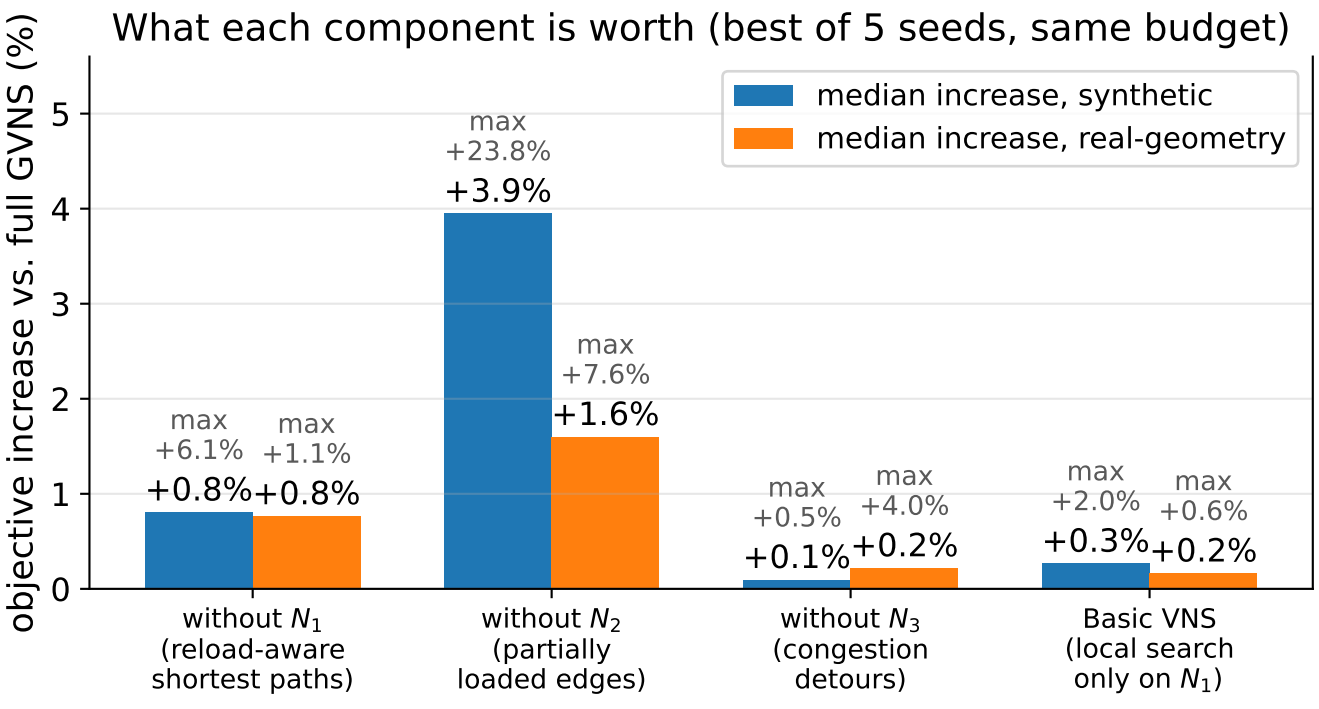
What each component is worth (best of 5 seeds, same budget)
objective increase vs. full GVNS (%)
0
1
2
3
4
5
median increase, synthetic
median increase, real-geometry
max +6.1%
+0.8%
max +1.1%
+0.8%
max +23.8%
+3.9%
max +7.6%
+1.6%
max +0.5%
+0.1%
max +4.0%
+0.2%
max +2.0%
+0.3%
max +0.6%
+0.2%
without N_1 (reload-aware shortest paths)
without N_2 (partially loaded edges)
without N_3 (congestion detours)
Basic VNS (local search only on N_1)